

Document 2 — Requirement Analysis

Incident Lifecycle Automation in ServiceNow

Project: Incident Lifecycle Automation in ServiceNow

College Project Submission

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1. Introduction

The Requirement Analysis phase is a fundamental step in the software development lifecycle. Its purpose is to identify and document the specific business, functional, technical, and operational needs that the proposed ServiceNow solution must fulfill. By clearly defining these requirements, we ensure that the final implementation aligns with organizational goals and user expectations.

This project aims to automate and standardize the IT Incident Management lifecycle within the ServiceNow platform. The scope encompasses the entire journey of an incident, including creation, classification, knowledge integration, assignment/escalation, investigation, change integration, management of child incidents, resolution, knowledge article creation, and comprehensive SLA tracking.

2. Business Requirements

The system must satisfy the following major business requirements to improve IT service delivery:

- **Standardized Incident Handling:** Provide a uniform method for creating and tracking incidents to ensure data consistency and accountability.
- **Improved Resolution Velocity:** Facilitate faster incident resolution through automated workflows and knowledge-based support.
- **Resource Optimization:** Ensure appropriate assignment to Level 2 support and efficient escalation paths to involve the right technical expertise.
- **Knowledge Management:** Support the reuse of solutions by integrating existing knowledge and enabling the creation of new articles from resolved incidents.
- **Integration and Relationship Management:** Allow for emergency change integration and the linking of child incidents to manage related or widespread issues effectively.
- **Performance Visibility:** Enable real-time monitoring of Service Level Agreements (SLAs) to maintain high service standards and compliance.

3. Functional Requirements

The following table outlines the specific functions the ServiceNow platform must perform:

ID	Functional Requirement
FR-01	The system should allow Service Desk agents to create incidents.
FR-02	The system should allow agents to classify incidents using appropriate fields (Category/Subcategory).

FR-03	The system should support Service and Service Offering selection.
FR-04	The system should support Configuration Item (CI) association.
FR-05	The system should provide Knowledge/Agent Assist support within the workspace.
FR-06	The system should allow incidents to be reassigned to appropriate support groups.
FR-07	The system should support Level 2 investigation and resolution workflows.
FR-08	The system should support emergency change integration when required.
FR-09	The system should allow child incidents to be created and linked to parent incidents.
FR-10	The system should allow cause and resolution information to be documented.
FR-11	The system should support Knowledge article creation and association from incidents.
FR-12	The system should provide SLA tracking through related records.

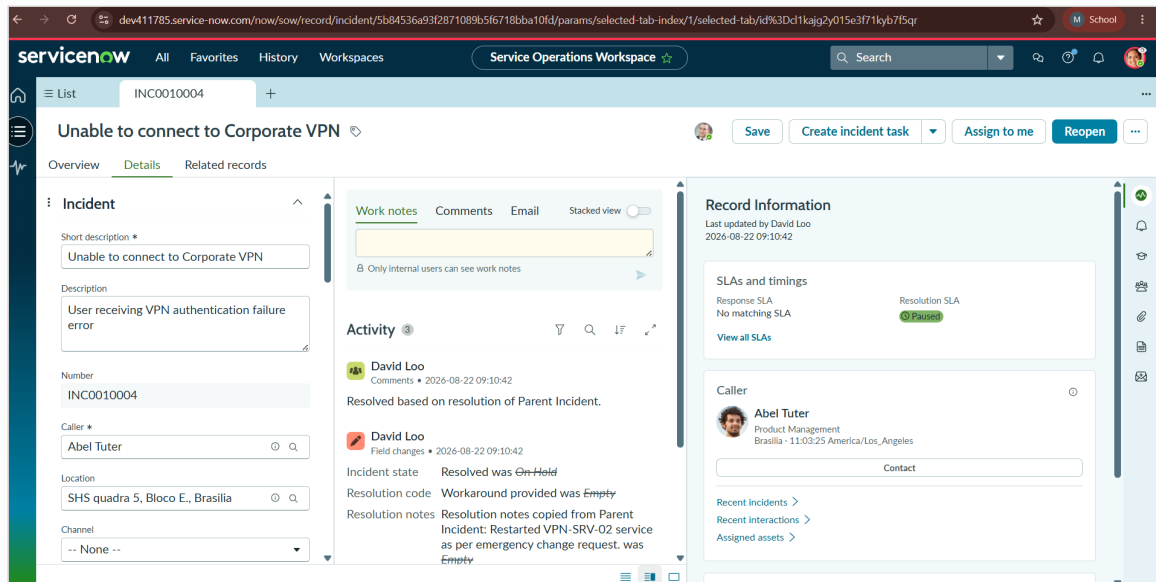


Figure 1: Incident creation capability in ServiceNow.

4. Non-Functional Requirements

Non-functional requirements define the quality attributes and constraints of the system:

Usability

The system should provide a structured and intuitive interface, specifically through the Service Operations Workspace, allowing agents to navigate the lifecycle efficiently.

Reliability

The platform must ensure that incident data, related records (Changes/Child Incidents), and SLA timers are consistently and accurately tracked throughout the lifecycle.

Maintainability

The solution should rely on standard, "out-of-the-box" ServiceNow capabilities to ensure that the process can be easily updated, patched, and monitored by administrators.

Scalability

The structured process must be capable of supporting a growing number of support teams and increasing incident volumes without performance degradation.

Visibility

Stakeholders should have clear visibility into the status, assignment, resolution progress, and SLA compliance of all incidents.

5. Functional Scope

The project scope is defined by the following ServiceNow capabilities and lifecycle stages:

- **Incident Record Management:** Creation, Classification, and CI/Service association.
- **Resolution Support:** Agent Assist and Knowledge Integration for faster troubleshooting.

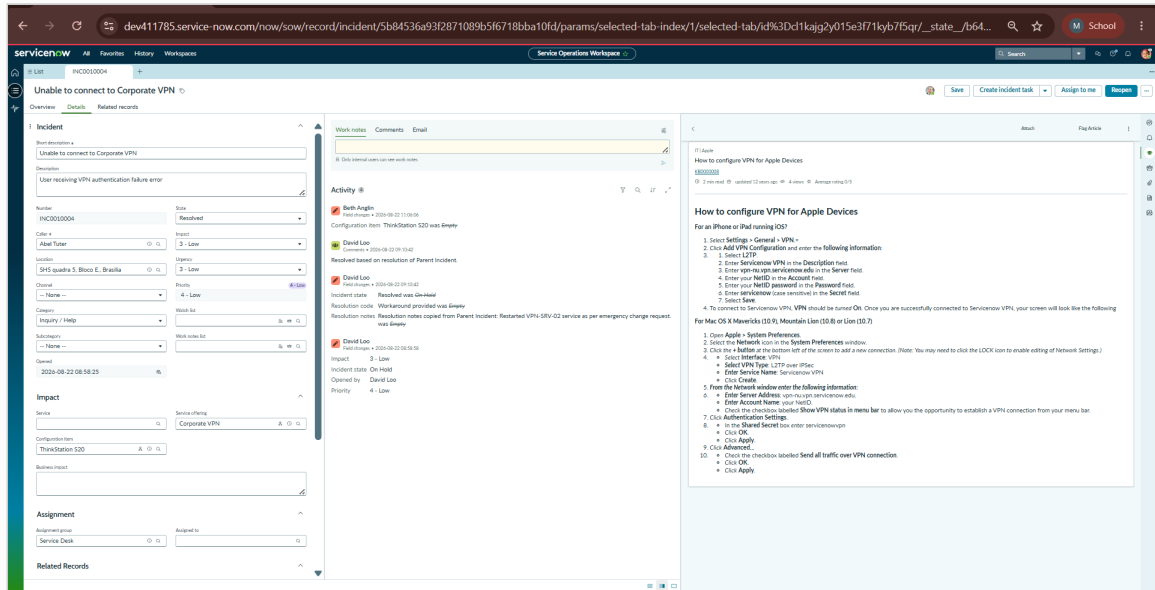


Figure 2: Agent Assist and Knowledge Integration capability.

- **Advanced Workflows:** Reassignment to L2, Emergency Change Integration, and Child Incident management.
- **Closure and Governance:** Documenting causes, creating knowledge articles, and SLA validation.

Proposed Lifecycle: Incident Creation → Classification → Knowledge Search → Assignment → Investigation → Change if Required → Resolution → Knowledge Creation → SLA Validation.

6. Stakeholder Mapping

Stakeholder	Role	Needs / Expectations	Impact of Automation
End Users	Report IT issues	Quick resolution, visibility	Faster resolution, transparent tracking
Service Desk Agents	First-level support	Easy logging, classification tools	Standardized intake, faster escalation
Level 2 Support	Diagnose and resolve	Complete data, related records	Better diagnosis, clear ownership
Change Management	Handle emergency changes	Structured change requests	Integrated change control
	Maintain config		Centralized monitoring

ServiceNow Admin		Scalable and trackable solution	
Incident Manager	Monitor lifecycle	SLA and escalation visibility	Better SLA compliance
Knowledge Manager	Manage knowledge	Accurate reusable solutions	Faster knowledge creation

7. User Roles

- **End User:** Reports issues and provides initial troubleshooting data.
- **Service Desk Agent:** Acts as the primary owner for logging, classifying, and initial triage.
- **Level 2 Support:** Provides deep technical analysis and permanent resolution for complex issues.
- **Change Management Team:** Approves and implements infrastructure changes linked to incidents.
- **Incident Manager:** Oversees the process to ensure SLAs are met and bottlenecks are removed.
- **Knowledge Manager:** Reviews and publishes knowledge created during the incident lifecycle.
- **ServiceNow Administrator:** Manages the platform's configuration and technical health.

8. System Requirements

The solution requires the following ServiceNow modules and capabilities:

- **Service Operations Workspace:** Central interface for modern incident management.
- **Agent Assist:** Contextual AI to provide solution recommendations.
- **Knowledge Management:** Repository for known errors and workarounds.
- **Change Management:** To facilitate infrastructure modifications.
- **SLA Engine:** To track Response and Resolution commitments.

9. Expected Outcomes

- Elimination of informal, untracked incident reporting.
- Improved Mean Time to Resolution (MTTR) through knowledge integration.
- Reduced human error in classification and assignment.
- Greater transparency for management via SLA tracking.

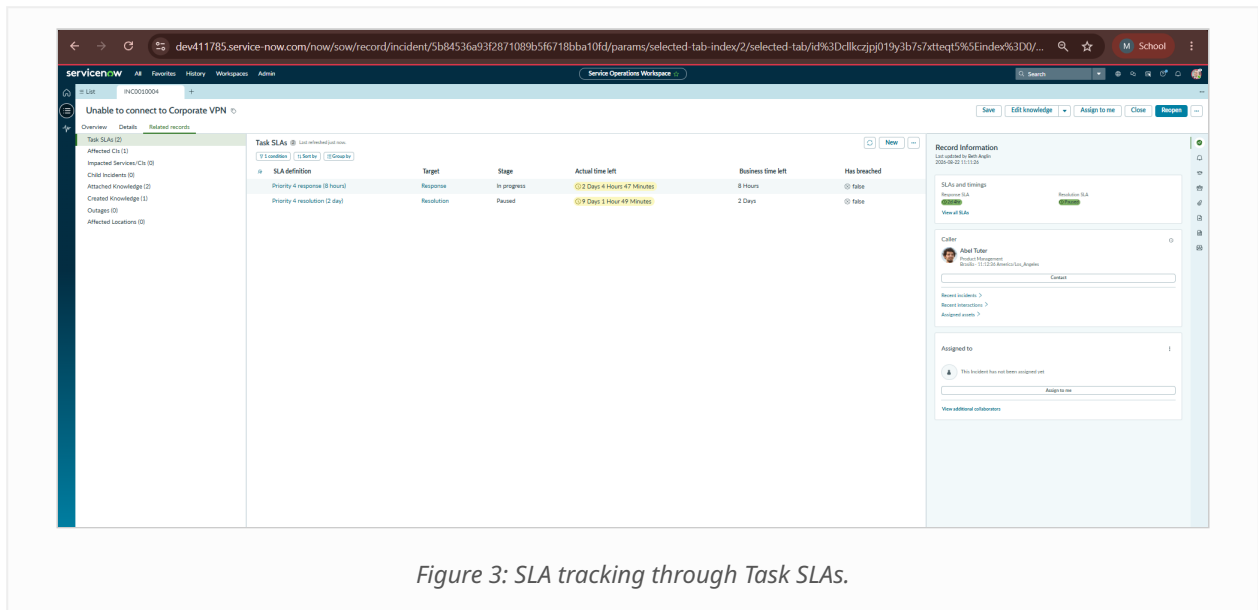


Figure 3: SLA tracking through Task SLAs.

10. Conclusion

The requirements outlined in this document provide a comprehensive blueprint for the Incident Lifecycle Automation project. By focusing on both the functional needs of the support teams and the strategic needs of the business, the identified ServiceNow capabilities will ensure a structured, reliable, and efficient incident management process.